```
import pandas as pd
df=pd.read_csv("medical_insurance.csv")
df.columns.tolist()
print(df.head(7))
print(df.shape)
print(df.info())
print(df.describe())
```

```
   age     sex     bmi  children smoker     region      charges
0   19  female  27.900         0    yes  southwest  16884.92400
1   18    male  33.770         1     no  southeast   1725.55230
2   28    male  33.000         3     no  southeast   4449.46200
3   33    male  22.705         0     no  northwest  21984.47061
4   32    male  28.880         0     no  northwest   3866.85520
5   31  female  25.740         0     no  southeast   3756.62160
6   46  female  33.440         1     no  southeast   8240.58960
(2772, 7)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2772 entries, 0 to 2771
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       2772 non-null   int64
 1   sex       2772 non-null   object
 2   bmi       2772 non-null   float64
 3   children  2772 non-null   int64
 4   smoker    2772 non-null   object
 5   region    2772 non-null   object
 6   charges   2772 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 151.7+ KB
None
               age          bmi     children       charges
count  2772.000000  2772.000000  2772.000000   2772.000000
mean     39.109668    30.701349     1.101732  13261.369959
std      14.081459     6.129449     1.214806  12151.768945
min      18.000000    15.960000     0.000000   1121.873900
25%      26.000000    26.220000     0.000000   4687.797000
50%      39.000000    30.447500     1.000000   9333.014350
75%      51.000000    34.770000     2.000000  16577.779500
max      64.000000    53.130000     5.000000  63770.428010
```

```
numerical_cols=df.select_dtypes(include=['int64','float64']).columns
print("NUMERICAL DATA")
print(list(numerical_cols))
print("\n")
categorical_cols=df.select_dtypes(include=['object']).columns
print("CATEGORICAL DATA")
print(list(categorical_cols))
```

```
NUMERICAL DATA
['age', 'bmi', 'children', 'charges']


CATEGORICAL DATA
['sex', 'smoker', 'region']
```

```
print(df.isnull().sum())
print(df.duplicated().sum())
```

```
age         0
sex         0
bmi         0
children    0
smoker      0
region      0
charges     0
dtype: int64
1435
```

```
df.drop_duplicates(inplace=True)
print(df.duplicated().sum())
print(df.shape)
```

```
0
(1337, 7)
```

Why Normalization is Required for the Selected Dataset

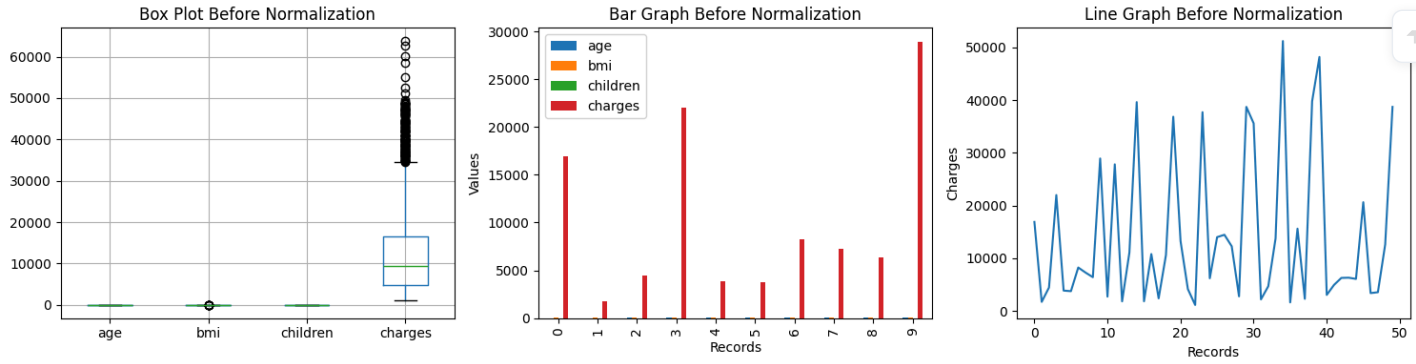
The Medical Insurance Cost Prediction dataset contains numerical attributes such as age, bmi, children, and charges. These attributes have different ranges of values. For example, age ranges from 18 to 64, while charges can range from a few thousand to several tens of thousands.

If the data is used without normalization, attributes with larger values may dominate those with smaller values, affecting the performance of machine learning algorithms. Normalization brings all numerical attributes to a common scale, ensuring that each feature contributes

equally to the analysis.

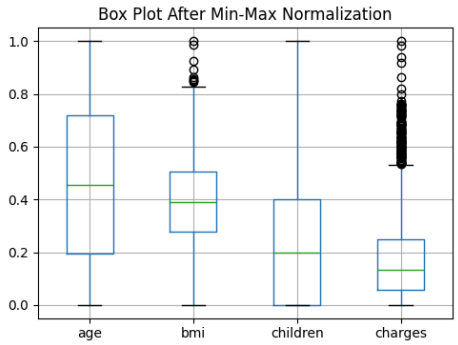
Normalization also improves model accuracy, speeds up the training process, and helps machine learning algorithms perform more effectively. Therefore, normalization is an important preprocessing step for the Medical Insurance Cost Prediction dataset.

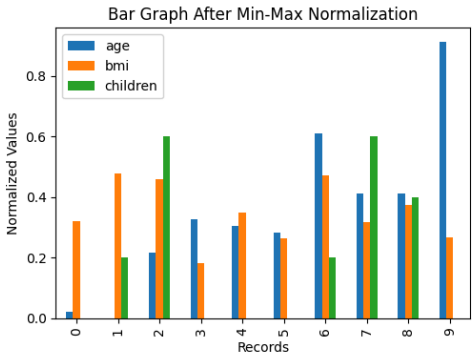
```
import matplotlib.pyplot as plt

plt.figure(figsize=(15,4))

# Box Plot
plt.subplot(1,3,1)
df[df.columns].boxplot()
plt.title("Box Plot Before Normalization")

# Bar Graph
plt.subplot(1,3,2)
df[df.columns].head(10).plot(kind='bar', ax=plt.gca())
plt.title("Bar Graph Before Normalization")
plt.xlabel("Records")
plt.ylabel("Values")

# Line Graph
plt.subplot(1,3,3)
plt.plot(df['charges'][:50])
plt.title("Line Graph Before Normalization")
plt.xlabel("Records")
plt.ylabel("Charges")
plt.tight_layout()
plt.show()
```

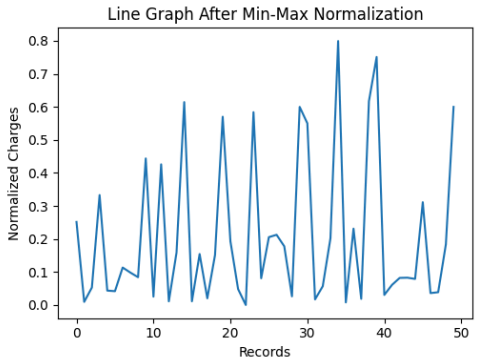



```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()

minmax_df = pd.DataFrame(
    scaler.fit_transform(df[['age','bmi','children','charges']]),
    columns=['age','bmi','children','charges']
)
```

```
import matplotlib.pyplot as plt

plt.figure(figsize=(15,4))

# Box Plot
plt.subplot(1,3,1)
minmax_df.boxplot()
plt.title("Box Plot After Min-Max Normalization")

# Bar Graph
plt.subplot(1,3,2)
minmax_df[['age','bmi','children']].head(10).plot(kind='bar', ax=plt.gca())
plt.title("Bar Graph After Min-Max Normalization")
plt.xlabel("Records")
plt.ylabel("Normalized Values")

# Line Graph
plt.subplot(1,3,3)
plt.plot(minmax_df['charges'][:50])
plt.title("Line Graph After Min-Max Normalization")
plt.xlabel("Records")
plt.ylabel("Normalized Charges")

plt.tight_layout()
plt.show()
```

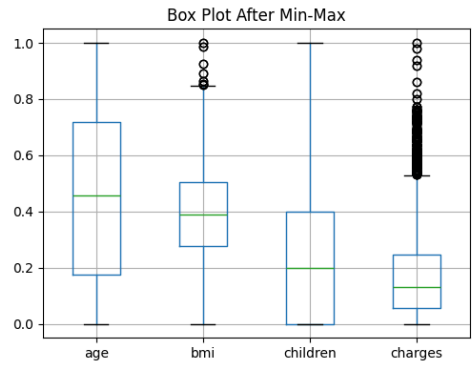


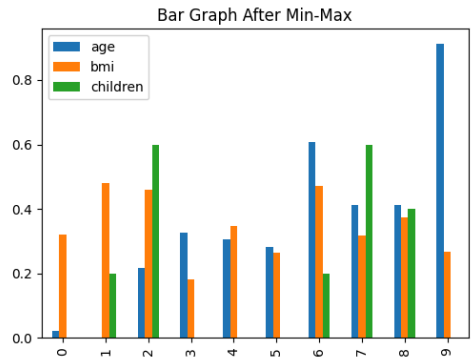


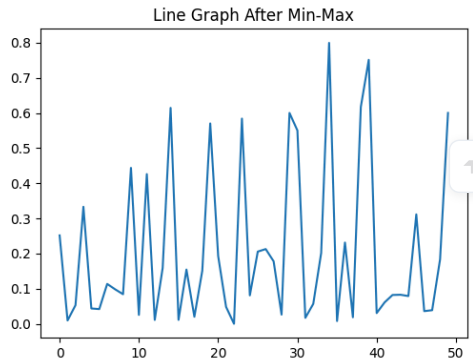



```
from sklearn.preprocessing import MinMaxScaler, StandardScaler
import pandas as pd
import matplotlib.pyplot as plt

# Numerical attributes
num_cols = ['age', 'bmi', 'children', 'charges']

# ================================================
# MIN-MAX NORMALIZATION
# ================================================

minmax_scaler = MinMaxScaler()

minmax_df = pd.DataFrame(
    minmax_scaler.fit_transform(df[num_cols]),
    columns=num_cols
)

print("MIN-MAX NORMALIZATION")
print(minmax_df.head())

print("\nORIGINAL VS MIN-MAX NORMALIZED VALUES")
comparison_minmax = pd.concat(
    [df[num_cols].head(), minmax_df.head()],
    axis=1,
    keys=['Original', 'Min-Max']
)
print(comparison_minmax)

# Visualization After Min-Max
plt.figure(figsize=(15,4))

plt.subplot(1,3,1)
minmax_df.boxplot()
plt.title("Box Plot After Min-Max")

plt.subplot(1,3,2)
minmax_df[['age','bmi','children']].head(10).plot(
    kind='bar',
    ax=plt.gca()
)
plt.title("Bar Graph After Min-Max")

plt.subplot(1,3,3)
plt.plot(minmax_df['charges'][:50])
plt.title("Line Graph After Min-Max")

plt.tight_layout()
plt.show()

# ================================================
# Z-SCORE NORMALIZATION
# ================================================

zscore_scaler = StandardScaler()

zscore_df = pd.DataFrame(
    zscore_scaler.fit_transform(df[num_cols]),
    columns=num_cols
)

comparison_zscore = pd.concat(
    [df[num_cols].head(), zscore_df.head()],
    axis=1,
    keys=['Original', 'Z-Score']
)
```

```
MIN-MAX NORMALIZATION
        age       bmi  children   charges
0  0.021739  0.321227       0.0  0.251611
1  0.000000  0.479150       0.2  0.009636
2  0.217391  0.458434       0.6  0.053115
3  0.326087  0.181464       0.0  0.333010
4  0.304348  0.347592       0.0  0.043816

ORIGINAL VS MIN-MAX NORMALIZED VALUES
  Original                                  Min-Max                    \
       age     bmi children      charges       age       bmi children
0       19  27.900        0  16884.92400  0.021739  0.321227      0.0
1       18  33.770        1   1725.55230  0.000000  0.479150      0.2
2       28  33.000        3   4449.46200  0.217391  0.458434      0.6
3       33  22.705        0  21984.47061  0.326087  0.181464      0.0
4       32  28.880        0   3866.85520  0.304348  0.347592      0.0


    charges
0  0.251611
1  0.009636
2  0.053115
3  0.333010
4  0.043816
```







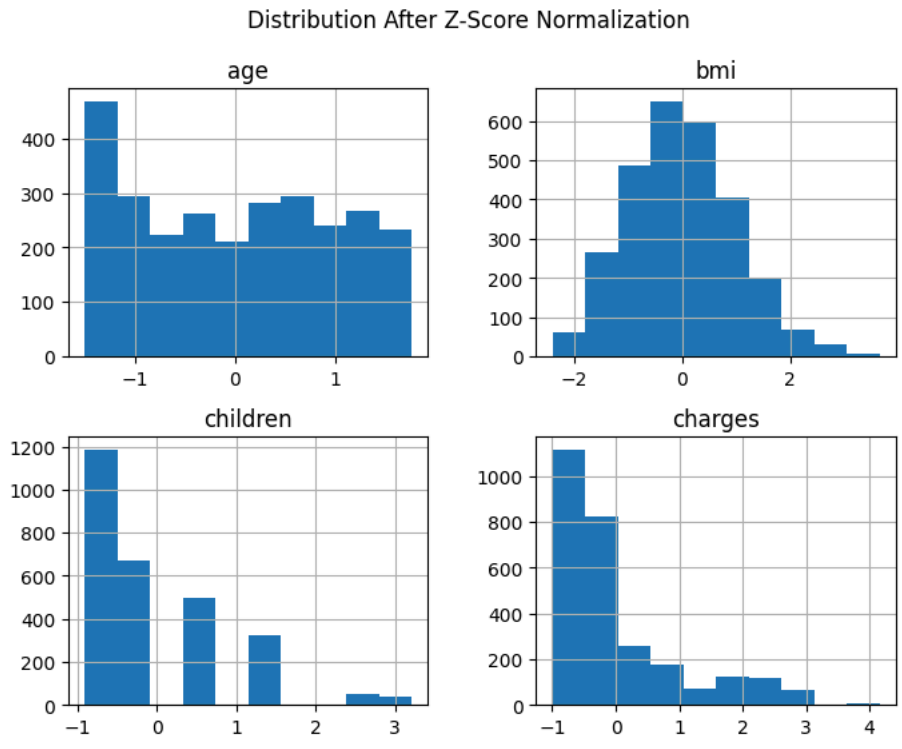

```
ORIGINAL VS Z-SCORE NORMALIZED VALUES
  Original                                  Z-Score                    \
       age     bmi children      charges       age       bmi  children
0       19  27.900        0  16884.92400 -1.428353 -0.457114 -0.907084
1       18  33.770        1   1725.55230 -1.499381  0.500731 -0.083758
2       28  33.000        3   4449.46200 -0.789099  0.375085  1.562893
3       33  22.705        0  21984.47061 -0.433959 -1.304814 -0.907084
4       32  28.880        0   3866.85520 -0.504987 -0.297201 -0.907084


    charges
0  0.298245
1 -0.949483
2 -0.725285
3  0.717976
4 -0.773238


Z-SCORE NORMALIZATION
        age       bmi  children   charges
0 -1.428353 -0.457114 -0.907084  0.298245
1 -1.499381  0.500731 -0.083758 -0.949483
2 -0.789099  0.375085  1.562893 -0.725285
3 -0.433959 -1.304814 -0.907084  0.717976
4 -0.504987 -0.297201 -0.907084 -0.773238

MEAN AFTER Z-SCORE
age         1.986546e-16
bmi        -6.190334e-16
children    3.588600e-17
charges     6.472296e-17
dtype: float64

STANDARD DEVIATION AFTER Z-SCORE
age         1.00018
bmi         1.00018
children    1.00018
charges     1.00018
dtype: float64
```

```
print("\nORIGINAL VS Z-SCORE NORMALIZED VALUES")
print(comparison_zscore)

print("\n\nZ-SCORE NORMALIZATION")
```

```
print(zscore_df.head())

print("\nMEAN AFTER Z-SCORE")
print(zscore_df.mean())

print("\nSTANDARD DEVIATION AFTER Z-SCORE")
print(zscore_df.std())
```

```
ORIGINAL VS Z-SCORE NORMALIZED VALUES
  Original                                Z-Score                      \
       age     bmi children      charges       age       bmi  children
0       19  27.900        0  16884.92400 -1.428353 -0.457114 -0.907084
1       18  33.770        1   1725.55230 -1.499381  0.500731 -0.083758
2       28  33.000        3   4449.46200 -0.789099  0.375085  1.562893
3       33  22.705        0  21984.47061 -0.433959 -1.304814 -0.907084
4       32  28.880        0   3866.85520 -0.504987 -0.297201 -0.907084


    charges
0  0.298245
1 -0.949483
2 -0.725285
3  0.717976
4 -0.773238


Z-SCORE NORMALIZATION
        age       bmi  children   charges
0 -1.428353 -0.457114 -0.907084  0.298245
1 -1.499381  0.500731 -0.083758 -0.949483
2 -0.789099  0.375085  1.562893 -0.725285
3 -0.433959 -1.304814 -0.907084  0.717976
4 -0.504987 -0.297201 -0.907084 -0.773238

MEAN AFTER Z-SCORE
age         1.986546e-16
bmi        -6.190334e-16
children    3.588600e-17
charges     6.472296e-17
dtype: float64

STANDARD DEVIATION AFTER Z-SCORE
age         1.00018
bmi         1.00018
children    1.00018
charges     1.00018
dtype: float64
```

```
# Distribution Analysis
zscore_df.hist(figsize=(8,6))
plt.suptitle("Distribution After Z-Score Normalization")
plt.show()
```



Comparison of Normalization Techniques

Min-Max Normalization: • Scales numerical values between 0 and 1. • Preserves the original relationships among data points. • Easy to interpret and suitable for algorithms that require bounded values. • Sensitive to outliers because extreme values affect the scaling range.

Z-Score Normalization: • Transforms data to have a mean of 0 and a standard deviation of 1. • Standardizes features with different scales. • Less sensitive to outliers compared to Min-Max Normalization. • Suitable for statistical analysis and machine learning algorithms.

Comparison: • Min-Max Normalization produces values in a fixed range (0 to 1). • Z-Score Normalization produces values centered around zero with no fixed range. • Min-Max is useful when the data has no significant outliers. • Z-Score is preferred when the dataset contains variations and extreme values.

For the Medical Insurance Cost Prediction dataset, Z-Score Normalization is the most suitable method.

Justification: • The charges attribute contains large variations and possible outliers. • Healthcare datasets often have features with different scales and distributions. • Z-Score Normalization standardizes the data using mean and standard deviation. • It reduces the effect of extreme values and improves the performance of machine learning models.

Conclusion: Both normalization techniques successfully transformed the numerical attributes. Min-Max Normalization scaled the values between 0 and 1, whereas Z-Score Normalization standardized the data to have a mean of 0 and a standard deviation of 1. For the Medical Insurance Cost Prediction dataset, Z-Score Normalization is more suitable because it handles variations and outliers more effectively.